

## 1 Week 9

Let  $m$  be a set function defined for all sets in a  $\sigma$ -algebra  $\mathcal{A}$  with values in  $[0, \infty]$ . Assume  $m$  is countably additive over countable disjoint collections of sets in  $\mathcal{A}$ .

**Question 1.1** (Royden 2.1). Prove that if  $A$  and  $B$  are two sets in  $\mathcal{A}$  with  $A \subseteq B$ , then  $m(A) \leq m(B)$ . This property is called *monotonicity*.

*Proof.* Since  $B = A \sqcup (B \setminus A)$  we have  $m(B) = m(A) + m(B \setminus A)$ . From this equality, it is evident that  $A$  may not have infinite measure unless  $B$  does as well, so the inequality  $m(A) \leq m(B)$  is obvious for when  $B$  has infinite measure. For  $B$  with finite measure, we have  $m(A) \leq m(A) + m(B \setminus A) = m(B)$  since  $m(B \setminus A) \geq 0$ .  $\square$

**Question 1.2** (Royden 2.2). Prove that if there is a set  $A$  in the collection  $\mathcal{A}$  for which  $m(A) < \infty$  then  $m(\emptyset) = 0$ .

*Proof.* Since  $\emptyset \subseteq A$ , we have by Royden 2.1 that  $m(\emptyset) \leq m(A)$  is finite. Since  $\emptyset = \emptyset \sqcup \emptyset$ , we have  $m(\emptyset) = m(\emptyset) + m(\emptyset) \Rightarrow m(\emptyset) = 0$ .  $\square$

**Question 1.3** (Royden 2.3). Let  $\{E_k\}_{k=1}^{\infty}$  be a countable collection of sets in  $\mathcal{A}$ . Prove that  $m(\cup_{k=1}^{\infty} E_k) \leq \sum_{k=1}^{\infty} m(E_k)$ .

*Proof.*  $\square$

**Question 1.4** (Royden 2.4). A set function  $c$ , defined on all subsets of  $\mathbb{R}$  is defined as follows. Define  $c(E)$  to be  $\infty$  if  $E$  has infinitely many members and  $c(E)$  to be the number of elements in  $E$  if  $E$  is finite; define  $c(\emptyset) = 0$ . Show that  $c$  is a countably additive and translation invariant set function. This set function is called the **counting measure**.

*Proof.*  $\square$

**Question 1.5** (Royden 2.8). Let  $B$  be the set of rational numbers in the interval  $[0, 1]$ , and let  $\{I_k\}_{k=1}^{\infty}$  be a finite collection of open intervals that covers  $B$ . Prove that  $\sum_{k=1}^n m^*(I_k) > 1$ .

*Proof.*  $\square$

**Question 1.6** (Royden 2.10). Let  $A$  and  $B$  be bounded sets for which there is an  $\alpha > 0$  such that  $|a - b| > \alpha$  for all  $a \in A, b \in B$ . Prove that  $m^*(A \cup B) = m^*(A) + m^*(B)$ .

*Proof.*  $\square$

**Question 1.7** (Royden 2.11). Prove that if a  $\sigma$ -algebra of subsets of  $\mathbb{R}$  contains intervals of the form  $(a, \infty)$ , then it contains all intervals.

*Proof.*  $\square$

**Question 1.8** (Royden 2.19). Let  $E$  have finite outer measure. Show that if  $E$  is not measurable, then there is an open set  $\mathcal{O}$  containing  $E$  that has finite outer measure and for which

$$m^*(\mathcal{O} \setminus E) > m^*(\mathcal{O}) - m^*(E).$$

*Proof.*  $\square$

**Question 1.9** (Royden 2.20: Lebesgue). Let  $E$  have finite outer measure. Show that  $E$  is measurable if and only if for each open, bounded interval  $(a, b)$ ,

$$b - a = m^*((a, b) \cap E) + m^*((a, b) \setminus E).$$

*Proof.*

□

**Question 1.10** (Royden 2.24). Show that if  $E_1$  and  $E_2$  are measurable, then

$$m(E_1 \cup E_2) + m(E_1 \cap E_2) = m(E_1) + m(E_2).$$

*Proof.*

□

**Question 1.11** (Royden 2.25). Show that the assumption that  $m(B_1) < \infty$  is necessary in part (ii) of the theorem regarding continuity of measure.

*Proof.*

□

**Question 1.12** (Royden 2.27). Let  $\mathcal{M}'$  be any  $\sigma$ -algebra of subsets of  $\mathbb{R}$  and  $m'$  a set function on  $\mathcal{M}'$  which takes values in  $[0, \infty]$ , is countably additive, and such that  $m'(\emptyset) = 0$ .

- (i) Show that  $m'$  is finitely additive, monotone, countably monotone, and possesses the excision property.
- (ii) Show that  $m'$  possesses the same continuity properties as Lebesgue measure.

*Proof.*

□

**Question 1.13** (Royden 2.28). Show that continuity of measure together with finite additivity of measure implies countable additivity of measure.

*Proof.*

□